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BÁO CÁO LAB 1

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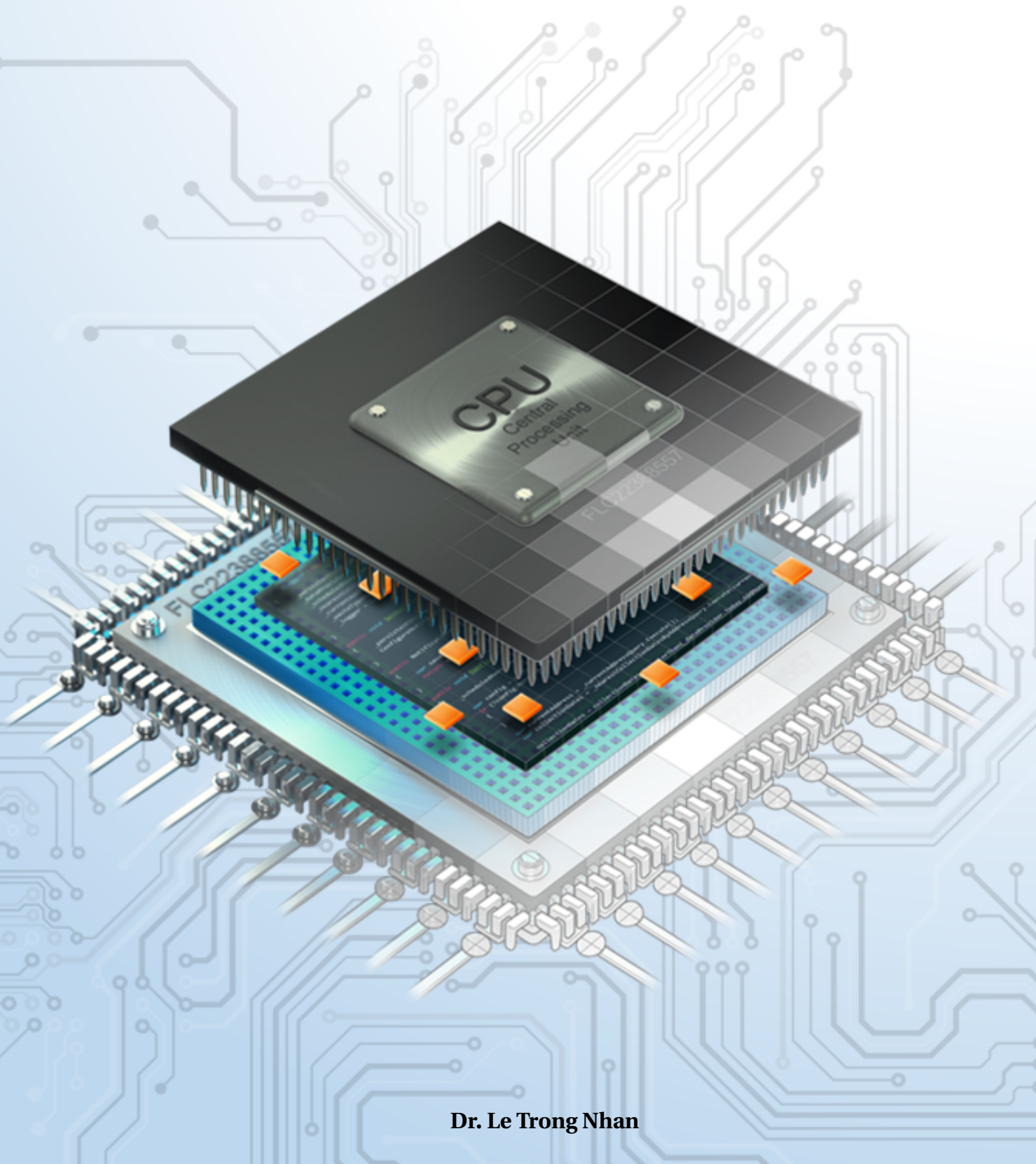
MSSV: 2312462

Thành phố Hồ Chí Minh, tháng 10 năm 2025



HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
COMPUTER ENGINEERING

Microcontroller



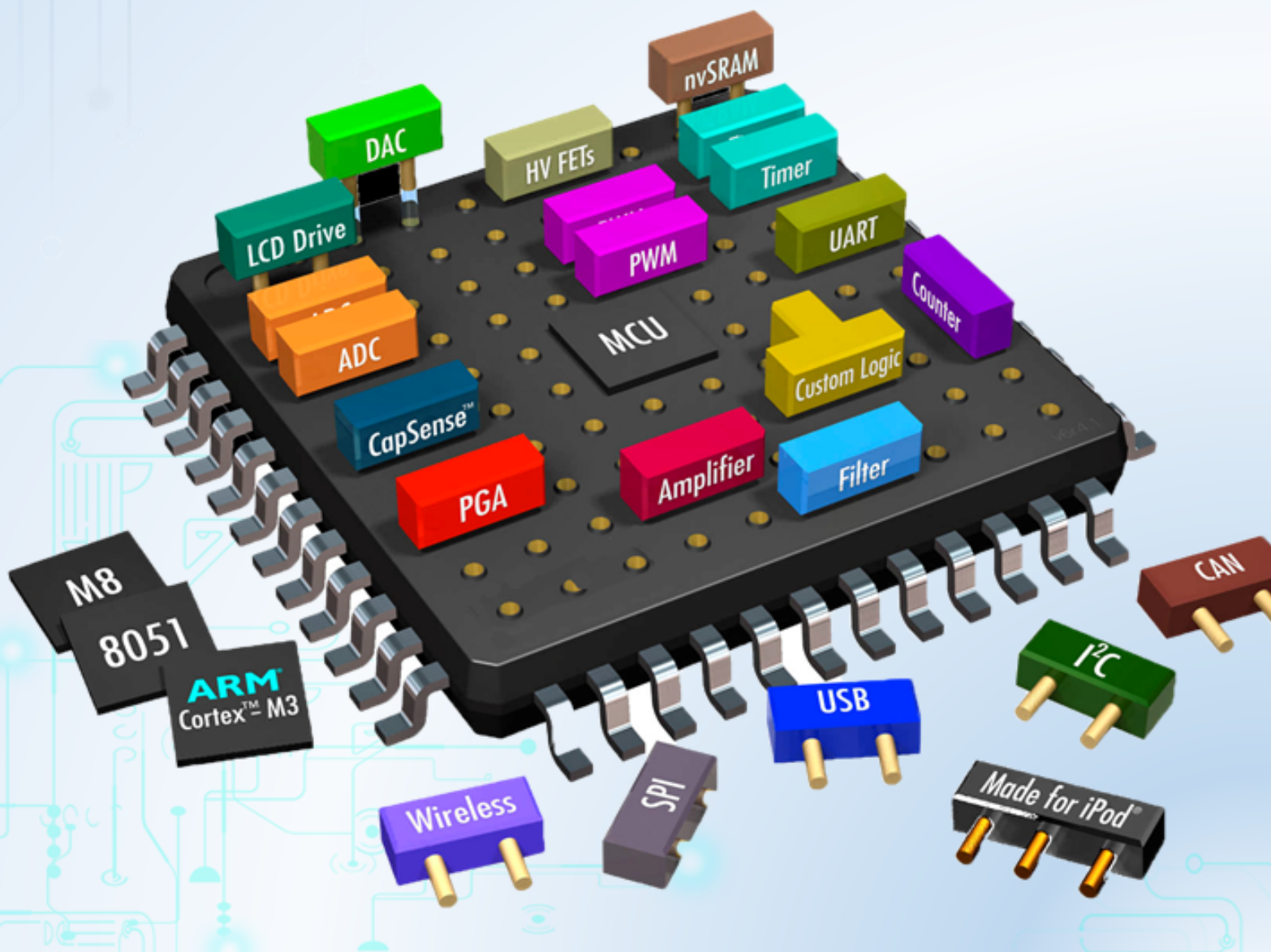
Dr. Le Trong Nhan

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CHƯƠNG 1

LED Animations



1 Introduction

In this manual, the STM32CubeIDE is used as an editor to program the ARM microcontroller. STM32CubeIDE is an advanced C/C++ development platform with peripheral configuration, code generation, code compilation, and debug features for STM32 microcontrollers and microprocessors.



Hình 1.1: STM32Cube IDE for STM32 Programming

The most interest of STM32CubeIDE is that after the selection of an empty STM32 MCU or MPU, or preconfigured microcontroller or microprocessor from the selection of a board, the initialization code generated automatically. At any time during the development, the user can return to the initialization and configuration of the peripherals or middleware and regenerate the initialization code with no impact on the user code. This feature can simplify the initialization process and speedup the development application running on STM32 micro-controller. The software can be downloaded from the link bellow:

https://ubc.sgp1.digitaloceanspaces.com/BKU_Softwares/STM32/stm32cubeide_1.7.0.zip

Moreover, for a hangout class, the program is firstly simulated on Proteus. Students are also supposed to download and install this software as well:

https://ubc.sgp1.digitaloceanspaces.com/BKU_Softwares/STM32/Proteus_8.10_SP0_Pro.exe

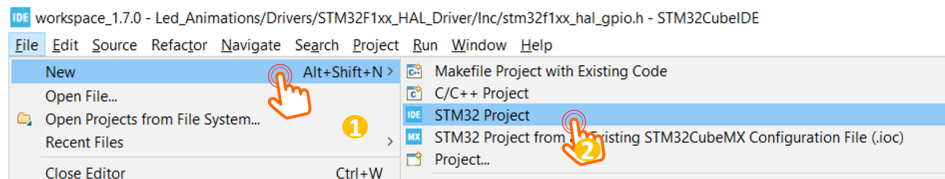
The rest of this manual consists of:

- Create a project on STM32Cube IDE
- Create a project on Proteus
- Simulate the project on Proteus

Finally, students are supposed to finish 10 different projects.

2 First project on STM32Cube

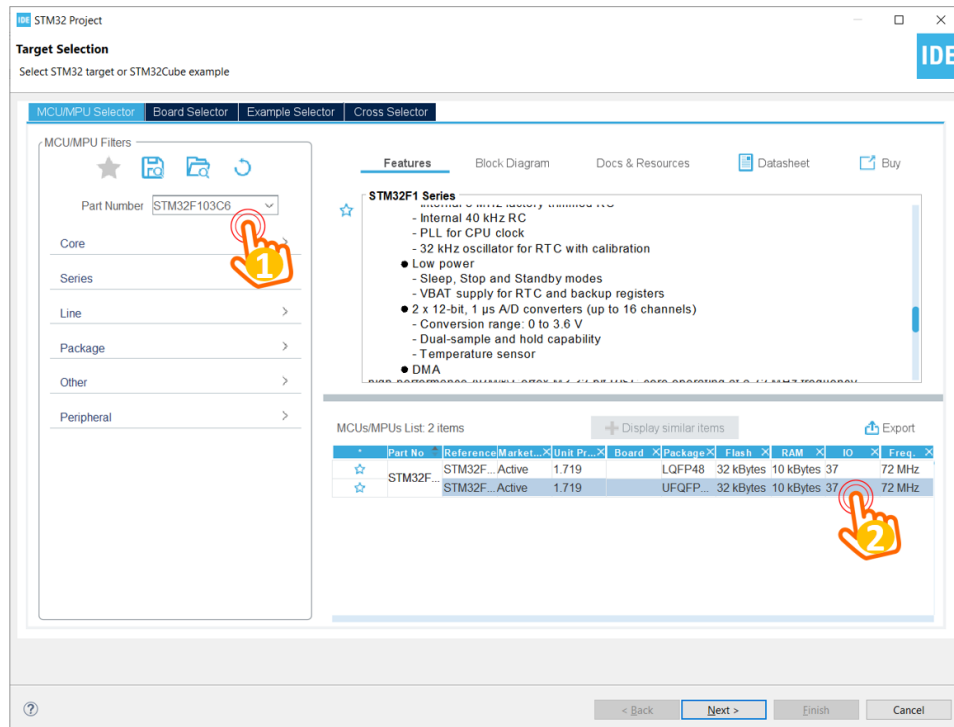
Step 1: Launch STM32CubeIDE, from the menu **File**, select **New**, then chose **STM32 Project**



Hình 1.2: Create a new project on STM32CubeIDE

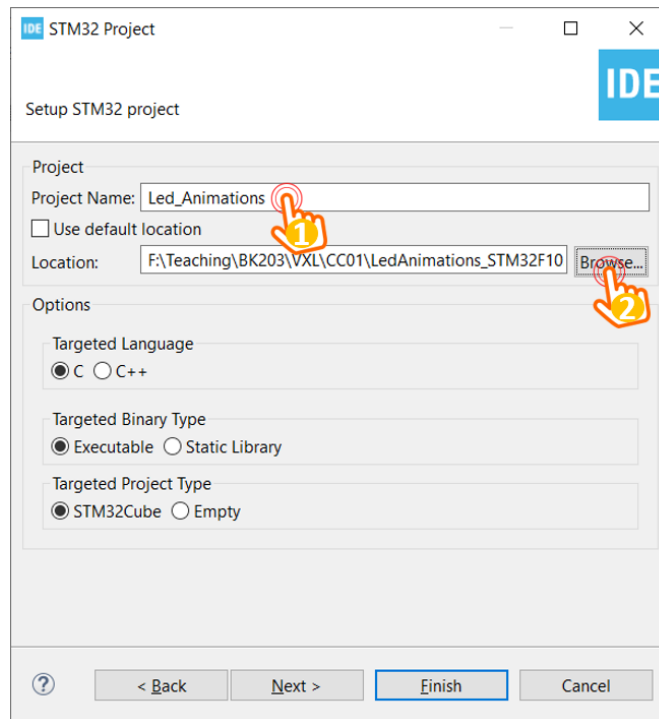
The IDE needs to download some packages, which normally takes time in this first time a project is created.

Step 2: Select the STM32F103C6 in the following dialog, then click on **Next**



Hình 1.3: Select the target device

Step 3: Provide the **Name** and the **Location** for the project.



Hình 1.4: Select the target device

It is important to notice that the **Targeted Project Type** should be **STM32Cube**. In the case this option is disabled, step 1 must be repeated. The location path should not contain special characters (e.g. the space). Finally, click on the **Next** button.

Step 4: On the last dialog, just keep the default firmware version and click on **Finish** button.

Step 5: The project is created and the wizard for configuration is displayed. This utility from CubeIDE can simplify the configuration process for an ARM microcontroller like the STM32.

From the configuration windows, select **Pin configuration**, select the pin **PA5** and set to **GPIO Output** mode, since this pin is connected to an LED in the STM32 development kit.

Step 6: Right click on PA5 and select **Enter user label**, and provide the name for this pin (e.g. **LED_RED**). This step helps programming afterward more memorable.

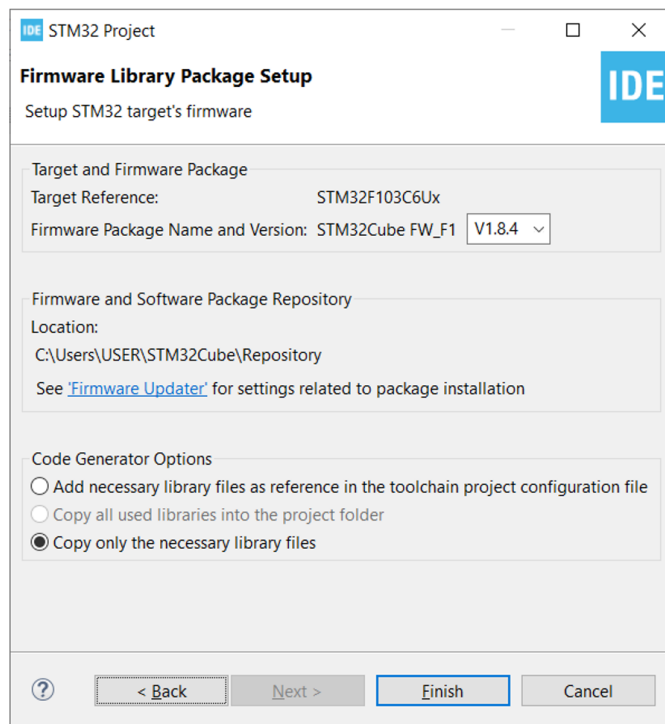
Finally, save the configuration process by pressing **Ctrl + S** and confirm this step by clicking on **OK** button. The code generation is started.

Step 7: Implement the first blinky project in the main function as follows:

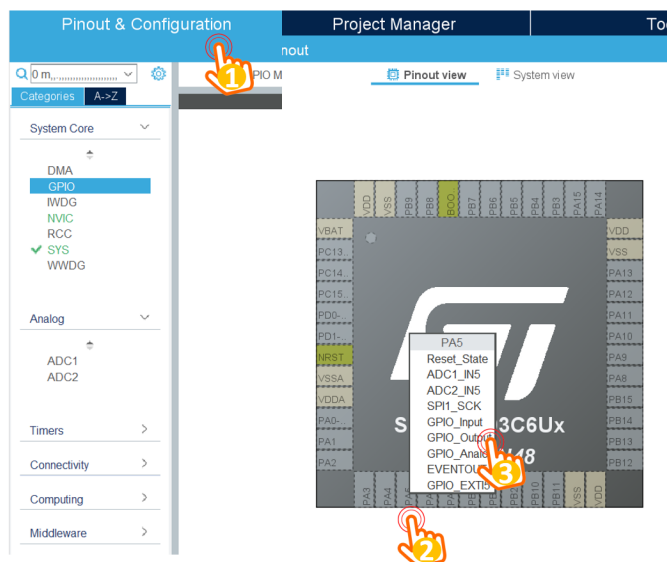
```

1 int main(void)
2 {
3     /* USER CODE BEGIN 1 */
4
5     /* USER CODE END 1 */

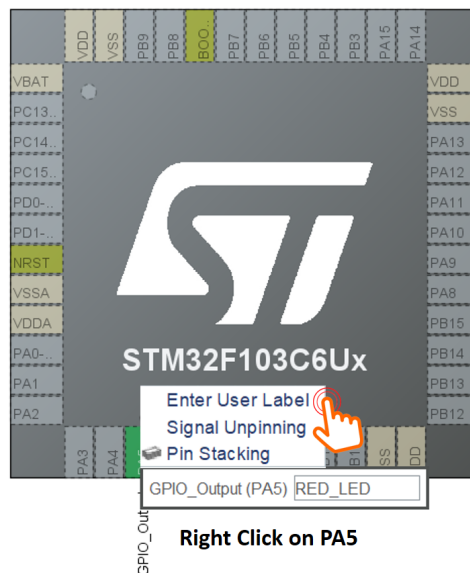
```



Hình 1.5: Keep default firmware version



Hình 1.6: Set PA5 to GPIO Output mode



Hình 1.7: Provide a name for PA5

```

6
7  /* MCU Configuration
8  -----
9  */
10
11 /* Reset of all peripherals, Initializes the Flash
12    interface and the Systick. */
13 HAL_Init();
14
15 /* USER CODE BEGIN Init */
16
17 /* USER CODE END Init */
18
19 /* Configure the system clock */
20 SystemClock_Config();
21
22 /* USER CODE BEGIN SysInit */
23
24 /* USER CODE END SysInit */
25
26 /* Initialize all configured peripherals */
27 MX_GPIO_Init();
28
29 /* USER CODE BEGIN 2 */
30
31 /* USER CODE END 2 */
32
33 /* Infinite loop */
34 /* USER CODE BEGIN WHILE */
35
36 while (1)
37 {

```

```

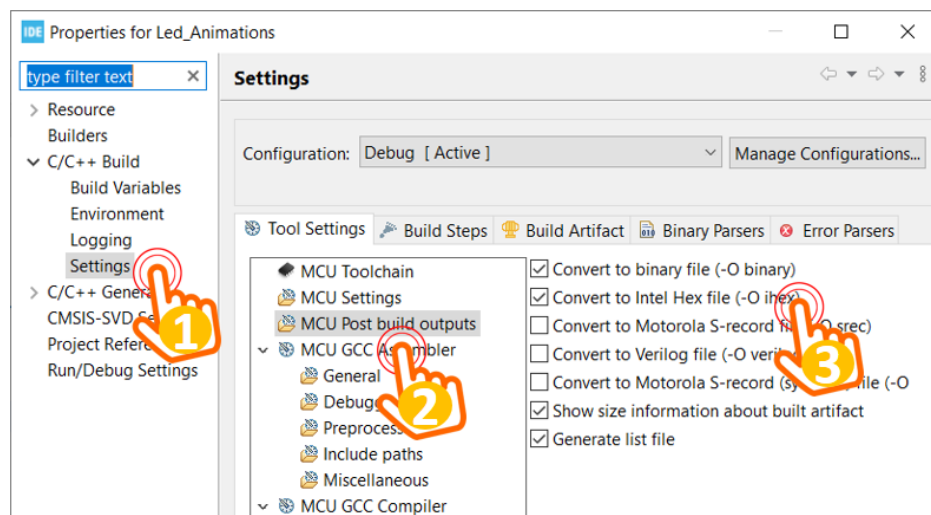
34     HAL_GPIO_TogglePin(LED_RED_GPIO_Port , LED_RED_Pin);
35     HAL_Delay(1000);
36     /* USER CODE END WHILE */
37
38     /* USER CODE BEGIN 3 */
39 }
40 /* USER CODE END 3 */
41 }

```

Program 1.1: First blinky LED project

Actually, what is added to the main function is line number 34 and 35. Please put your code in a right place, otherwise it can be deleted when the code is generated (e.g. change the configuration of the project). When coding, frequently use the suggestions by pressing **Ctrl+Space**.

Step 7: Due to the simulation on Proteus, the hex file should be generated from STM32Cube IDE. From menu **Project**, select **Properties** to open the dialog bellow:



Hình 1.8: Config for hex file output

Navigate to **C/C++ Build**, select **Settings**, **MCU Post build outputs**, and check to the **Intel Hex file**.

Step 8: Build the project by clicking on menu **Project** and select **Build Project**. Please check on the output console of the IDE to be sure that the hex file is generated, as follow:

```

22:36:06 **** Incremental Build of configuration Debug for project Led_Animations ****
make -j8 all
arm-none-eabi-size  Led_Animations.elf
   text    data    bss     dec     hex filename
   4596     20    1572    6188    182c Led_Animations.elf
Finished building: default.size.stdout

22:36:06 Build Finished. 0 errors, 0 warnings. (took 272ms)

```

Hình 1.9: Compile the project and generate Hex file

The hex file is located under the **Debug** folder of your project, which is used for the simulation in Proteus afterward. In the case a development kit is connected to your PC, from menu **Run**, select **Run** to download the program to the hardware platform.

In the case there are multiple project in a work-space, double click on the project name to activate this project. Whenever a project is built, check the output files to make sure that you are working in a right project.

3 Simulation on Proteus

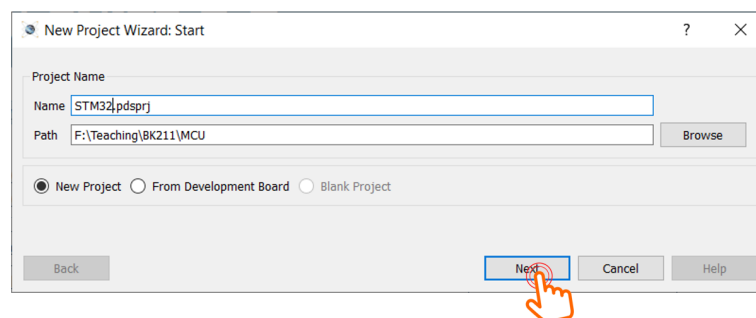
For an online training, a simulation on Proteus can be used. The details to create an STM32 project on Proteus are described bellow.

Step 1: Launch Proteus (with administration access) and from menu **File**, select **New Project**.



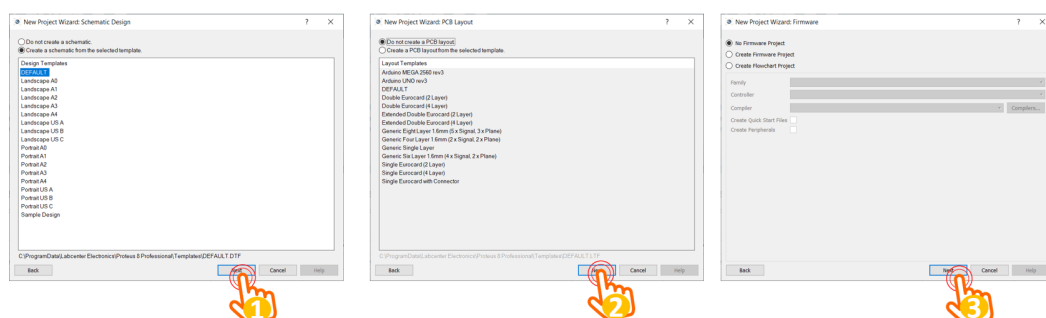
Hình 1.10: Create a new project on Proteus

Step 2: Provide the name and the location of the project, then click on **Next** button.



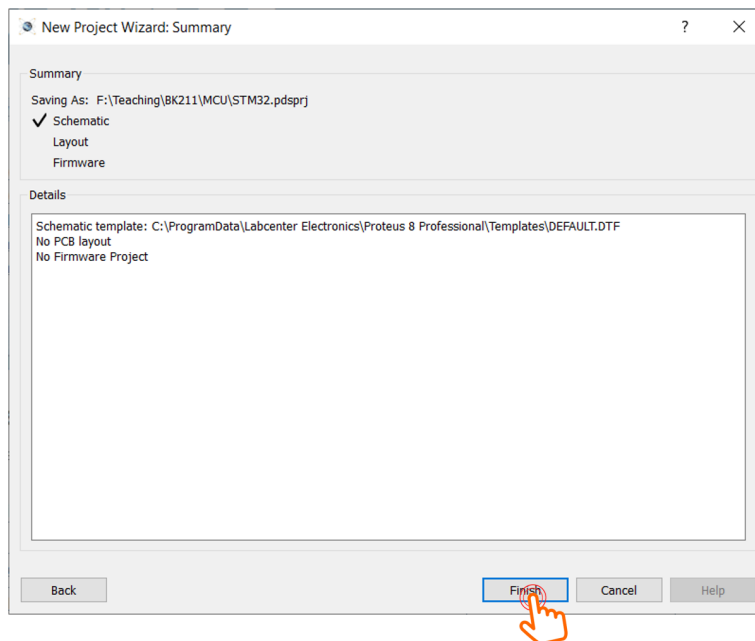
Hình 1.11: Provide project name and location

Step 3: For following dialog, just click on **Next** button as just a schematic is required for the lab.



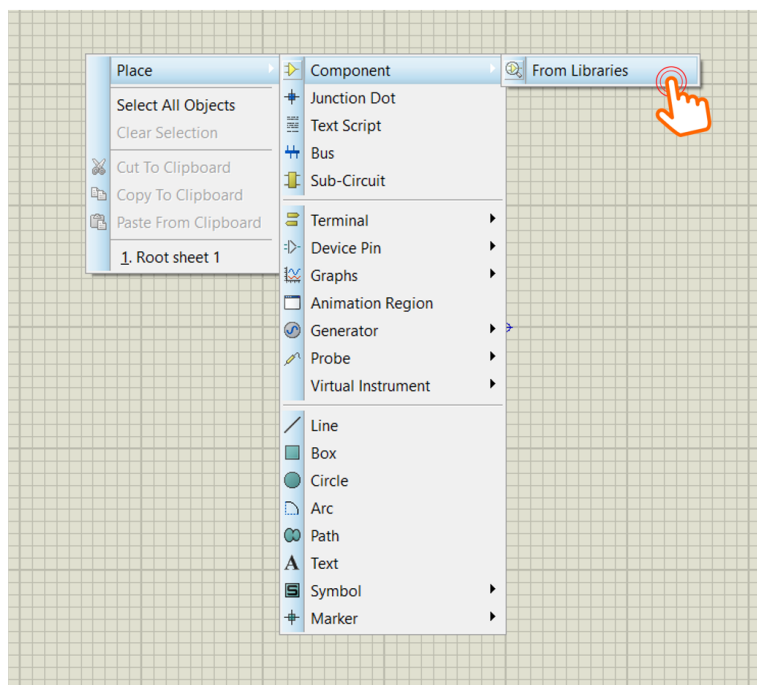
Hình 1.12: Keep the default options by clicking on Next

Step 4: Finally, click on **Finish** button to close the project wizard.



Hình 1.13: Finish the project wizard

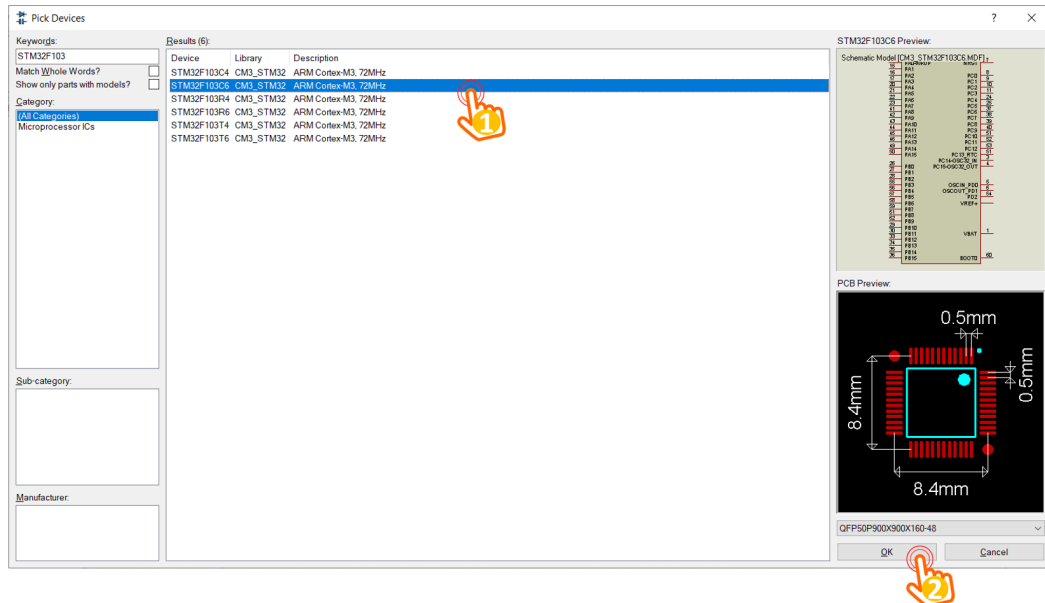
Step 5: On the main page of the project, right click to select **Place, Components, From Libraries**, as follows:



Hình 1.14: Select a component from the library

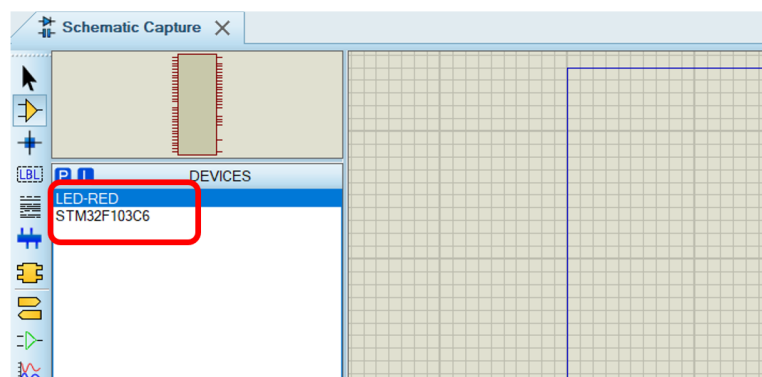
If there is an error with no library found, please restart the Proteus software with Run as administrator option.

Step 6: From the list of components in the library, select STM32F103C6, as follows:



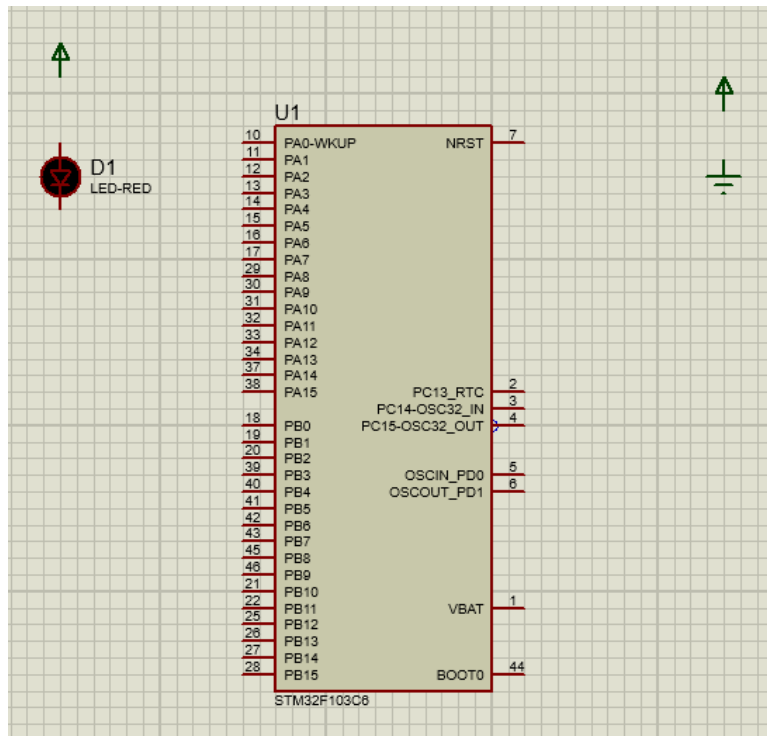
Hình 1.15: Select STM32F103C6

Repeat step 5 and 6 to select an LED, named **LED-RED** in Proteus. Finally, these components are appeared on the DEVICES windows, which is on left hand side as follows:



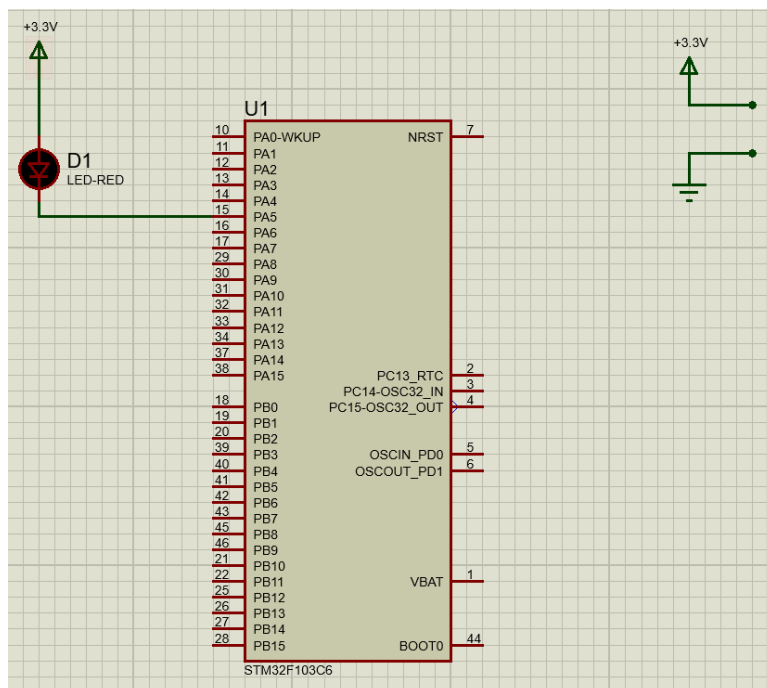
Hình 1.16: STM32 and an LED in the project

Step 7: Place the components to the project: right click on the main page, select on **Place, Component**, and select device added in Step 6. To add the Power and the Ground, right click on the main page, select on **Place, Terminal**. The result in this step is expected as follows:



Hình 1.17: Place components to the project

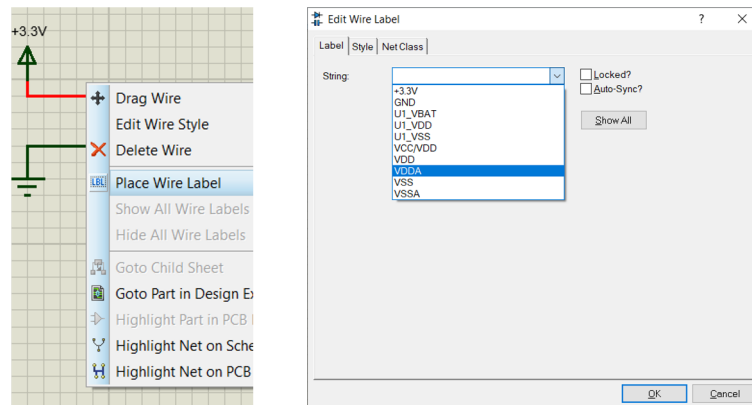
Step 8: Start wiring the circuit. The negative pin of the LED is connected to PA5 while its positive pin is connected to the power supply. For the power and the ground on the right, just make a short wire, which will be labeled in the next step.



Hình 1.18: Connect components and set the power to 3.3V

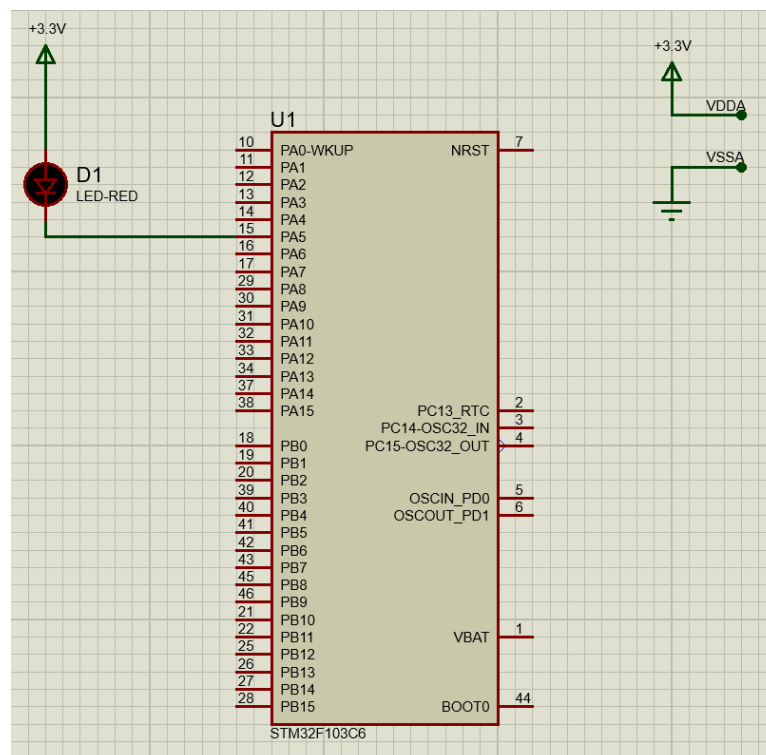
In this step, also double click on the power supply in order to provide the String property to **+3.3V**.

Step 8: Right click on the wire of the power supply and the ground, and select **Place wire Label**



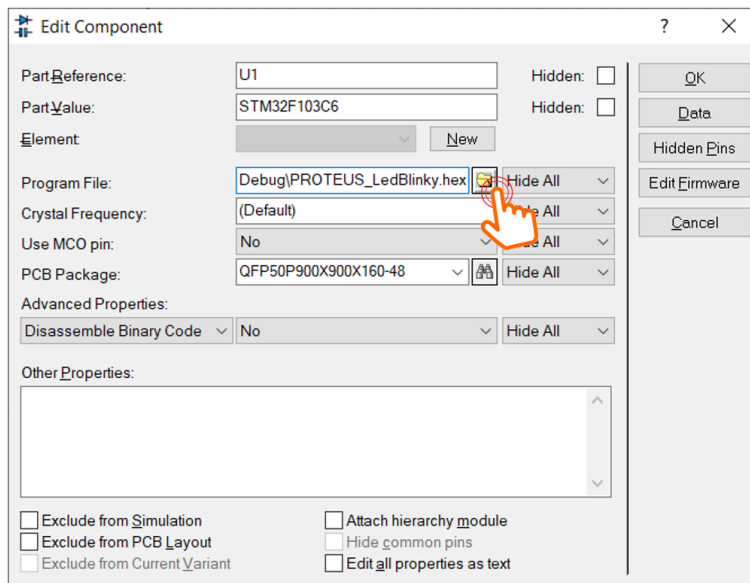
Hình 1.19: Place label for Power and Ground

This step is required as VDDA and VSSA of the STM32 must be connected to provide the reference voltage. Therefore, VDDA is connected to 3.3V, while the VSSA is connected to the Ground. Finally, the image of our schematic is shown below:



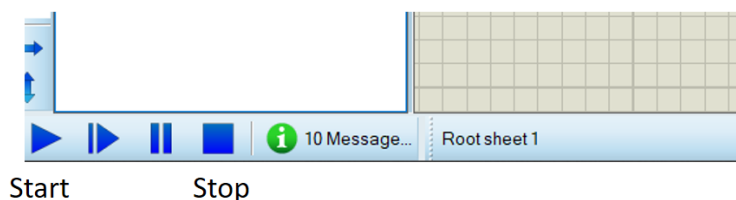
Hình 1.20: Finalize the schematic

Step 9: Double click on the STM32, and set the **Program File** to the Hex file, which is generated from Cube IDE, as following:



Hình 1.21: Set the program of the STM32 to the hex file from Cube IDE

From now, the simulation is ready to start by clicking on the menu **Debug**, and select on **Run simulation**. To stop the simulation, click on **Debug** and select **Stop VMS Debugging**. Moreover, there are some quick access bottom on the left corner of the Proteus to start or stop the simulation, as shown following:



Hình 1.22: Quick access buttons to start and stop the simulation

If everything is success, students can see the LED is blinking every second. Please stop the simulation before updating the project, either in Proteus or STM32Cube IDE. However, the step 9 (set the program file for STM32 in Proteus) is required to do once. Beside the toggle instruction, student can set or reset a pin as following:

```
1 while (1){
2     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin,
3       GPIO_PIN_SET);
4     HAL_Delay(1000);
5     HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin,
6       GPIO_PIN_RESET);
7     HAL_Delay(1000);
8 }
```

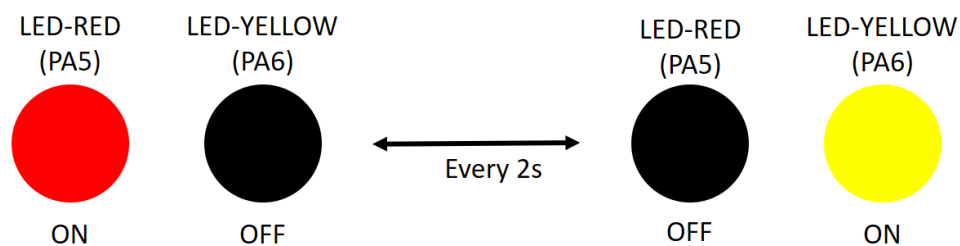
Program 1.2: An example for LED blinky

4 Exercise and Report

4.1 Exercise 1

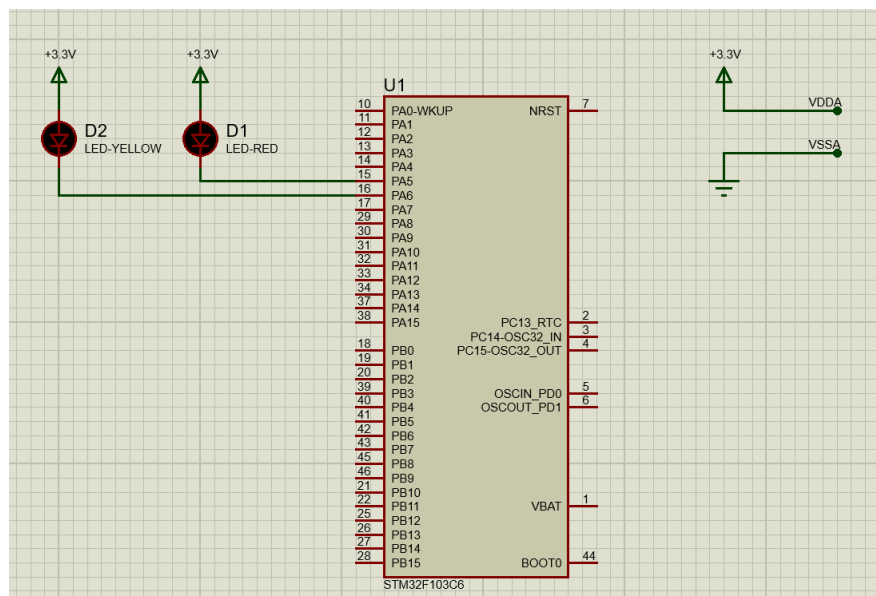
From the simulation on Proteus, one more LED is connected to pin **PA6** of the STM32 (negative pin of the LED is connected to PA6). The component suggested in this exercise is **LED-YELLOW**, which can be found from the device list.

In this exercise, the status of two LEDs are switched every 2 seconds, as demonstrated in the figure bellow.



Hình 1.23: State transitions for 2 LEDs

Report 1: Depict the schematic from Proteus simulation in this report. The caption of the figure is a downloadable link to the Proteus project file (e.g. a github link).



Hình 1.24: [Exersice 1 Proteus project file github link](#)

Report 2: Present the source code in the infinite loop while of your project. If a user-defined functions is used, it is required to present in this part. A brief description can be added for this function (e.g. using comments). A template to present your source code is presented bellow.

```

1 HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, RESET);
2 HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin,
3   SET);
4 while (1)
5 {
6     HAL_Delay(2000);
7     HAL_GPIO_TogglePin(LED_RED_GPIO_Port, LED_RED_Pin);
8     HAL_GPIO_TogglePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
9 }

```

Program 1.3: [Exercise 1 Source github link](#)

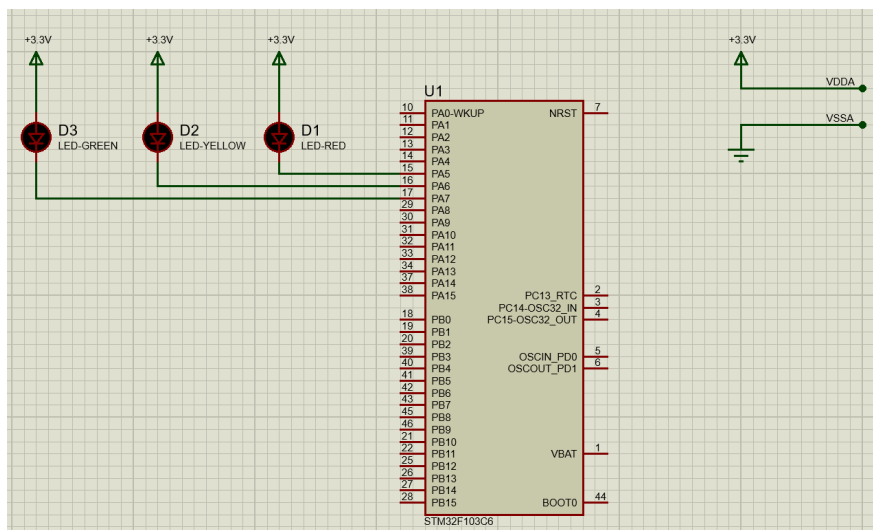
Demo video: [Exercise 1 Demo video link](#) (beginning at 00:00).

4.2 Exercise 2

Extend the first exercise to simulate the behavior of a traffic light. A third LED, named **LED-GREEN** is added to the system, which is connected to **PA7**. A cycle in this traffic light is 5 seconds for the RED, 2 seconds for the YELLOW and 3 seconds for the GREEN. The LED-GREEN is also controlled by its negative pin.

Similarly, the report in this exercise includes the schematic of your circuit and a your source code in the while loop.

Report 1: Present the schematic.



Hình 1.25: [Exercise 2 Proteus project file github link](#)

Report 2: Present the source code in while.

```

1 // LED active-low: RESET = ON, SET = OFF
2 static inline void led_on(GPIO_TypeDef* port, uint16_t pin)
3 {
4     HAL_GPIO_WritePin(port, pin, GPIO_PIN_RESET);
5 }
6 static inline void led_off(GPIO_TypeDef* port,
7     uint16_t pin)
8 {
9     HAL_GPIO_WritePin(port, pin, GPIO_PIN_SET);
10 }
11 // Init State: GREEN
12 static void traffic_light_init(void) {
13     led_off(LED_RED_GPIO_Port, LED_RED_Pin);
14     led_off(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
15     led_on (LED_GREEN_GPIO_Port, LED_GREEN_Pin);
16 }
17 // One loop of traffic-light state machine: set LEDs, wait,
18 //next state (GREEN -> YELLOW -> RED -> GREEN)
19 static void traffic_light_loop_once(void) {
20     static traffic_state_t state = STATE_GREEN;
21     uint32_t delay_ms = 0;
22
23     switch (state) {
24         case STATE_GREEN: // GREEN 3s -> YELLOW
25             led_off(LED_RED_GPIO_Port, LED_RED_Pin);
26             led_off(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
27             led_on (LED_GREEN_GPIO_Port, LED_GREEN_Pin);
28             delay_ms = time_green_ms;
29             state = STATE_YELLOW;
30             break;
31
32         case STATE_YELLOW: // YELLOW 2s -> RED
33             led_off(LED_RED_GPIO_Port, LED_RED_Pin);
34             led_on (LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
35             led_off(LED_GREEN_GPIO_Port, LED_GREEN_Pin);
36             delay_ms = time_yellow_ms;
37             state = STATE_RED;
38             break;
39
40         case STATE_RED: // RED 5s -> GREEN
41             led_on (LED_RED_GPIO_Port, LED_RED_Pin);
42             led_off(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
43             led_off(LED_GREEN_GPIO_Port, LED_GREEN_Pin);
44             delay_ms = time_red_ms;
45             state = STATE_GREEN;
46             break;
47     }
48     HAL_Delay(delay_ms);
49 }

```

```

50 while (1)
51 {
52     traffic_light_loop_once();
53 }

```

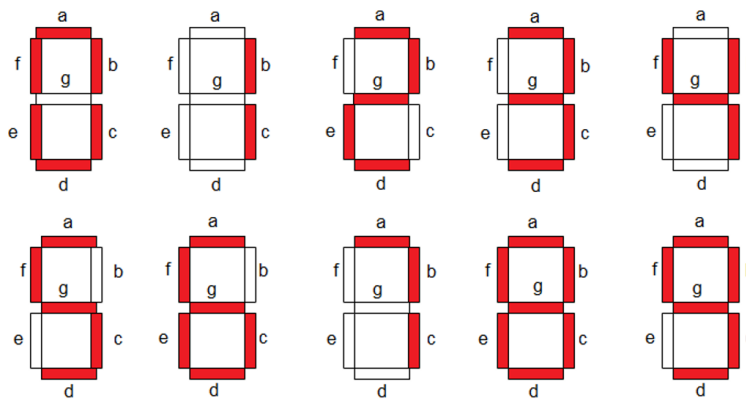
Program 1.4: [Exercise 2 Source github link](#)

Demo video: [Exercise 2 Demo video link](#) (beginning at 00:28).

4.3 Exercise 4

Add **only one 7 led segment** to the schematic in Exercise 3. This component can be found in Proteus by the keyword **7SEG-COM-ANODE**. For this device, the common pin should be connected to the power supply and other pins are supposed to be connected to PB0 to PB6. Therefore, to turn-on a segment in this 7SEG, the STM32 pin should be in logic 0 (0V).

Implement a function named **display7SEG(int num)**. The input for this function is from 0 to 9 and the outputs are listed as following:



Hình 1.26: Display a number on 7 segment LED

This function is invoked in the while loop for testing as following:

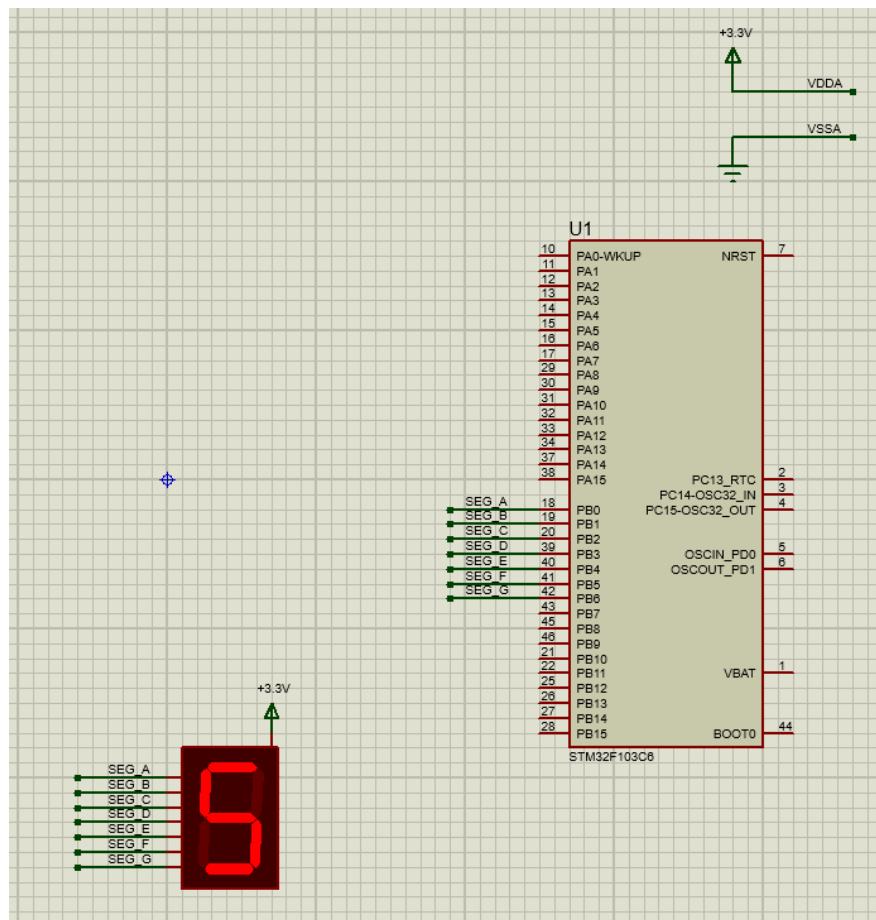
```

1 int counter = 0;
2 while (1){
3     if(counter >= 10) counter = 0;
4     display7SEG(counter++);
5     HAL_Delay(1000);
6
7 }

```

Program 1.5: An example for your source code

Report 1: Present the schematic.



Hình 1.27: Exercice 4 Proteus project file [github link](#)

Report 2: Present the source code for display7SEG function.

```

1 // Show a number 0-9 on 7-segment
2 static void display7SEG(int num) {
3     static const uint8_t seg_code[10] = {
4         0xC0, // 0 -> 1100 0000 : a b c d e f = 0 (ON), g =
5         1 (OFF)
6         0xF9, // 1 -> 1111 1001 : b c = ON
7         0xA4, // 2 -> 1010 0100 : a b g e d = ON
8         0xB0, // 3 -> 1011 0000 : a b c d g = ON
9
10    };
11
12    for (int i = 0; i < 7; i++) {
13        digitalWrite(seg_code[num] & (1 << i), HIGH);
14    }
15
16    digitalWrite(7, LOW);
17 }

```

```

8      0x99, // 4 -> 1001 1001 : b c f g = ON
9      0x92, // 5 -> 1001 0010 : a c d f g = ON
10     0x82, // 6 -> 1000 0010 : a c d e f g = ON
11     0xF8, // 7 -> 1111 1000 : a b c = ON
12     0x80, // 8 -> 1000 0000 : all segments ON
13     0x90  // 9 -> 1001 0000 : a b c d f g = ON
14 };
15
16     if (num < 0 || num > 9) return;
17
18     uint8_t pattern = seg_code[num];
19
20     HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, (pattern
21 & (1<<0)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
22     HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, (pattern
23 & (1<<1)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
24     HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, (pattern
25 & (1<<2)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
26     HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, (pattern
27 & (1<<3)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
28     HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, (pattern
29 & (1<<4)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
30     HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, (pattern
31 & (1<<5)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
32     HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, (pattern
33 & (1<<6)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
34
35     // Example:
36     // - For digit 0 (num = 0): pattern = 0xC0 = 1100 0000
37     //   ; (1<<0) = 0000 0001
38     //   bit0 = 0 -> (pattern & (1<<0)) == 0000 0000 = 0
39     //   -> write GPIO_PIN_RESET -> segment A turns ON
40     // - For digit 1: pattern = 0xF9 = 1111 1001 ; (1<<0) =
41     //   0000 0001
42     //   bit0 = 1 -> (pattern & (1<<0)) != 0000 0001 ->
43     //   write GPIO_PIN_SET -> segment A stays OFF
44 }
45
46 int counter = 0;
47 while (1){
48     if(counter >= 10) counter = 0;
49     display7SEG(counter++);
50     HAL_Delay(1000);
51 }

```

Program 1.6: [Exercise 4 Source github link](#)

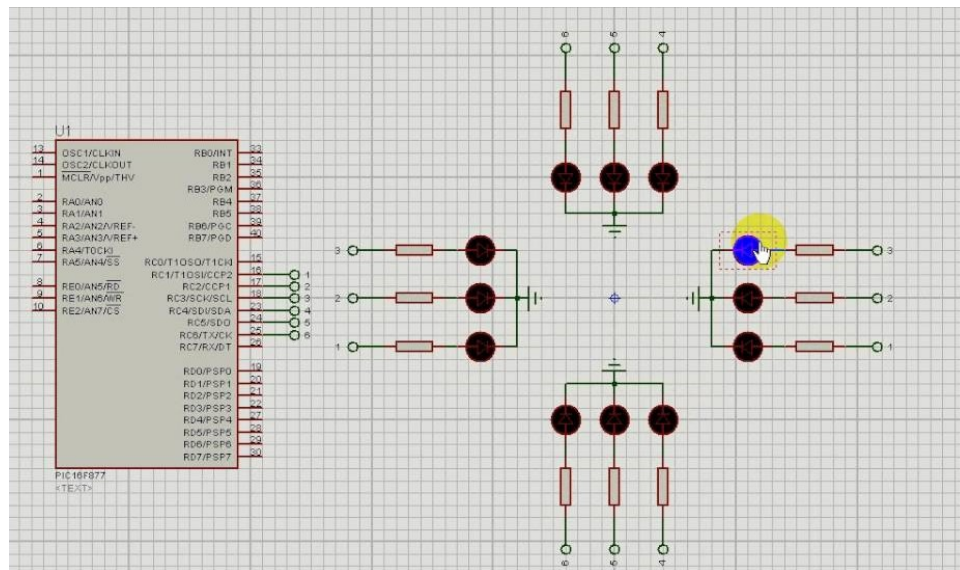
Demo video: [Exercise 4 Demo video link](#) (beginning at 01:42).

4.4 Exercise 3 & Exercise 5

Extend to the 4-way traffic light. Arrange 12 LEDs in a nice shape to simulate the behaviors of a traffic light. A reference design can be found in the figure bellow.

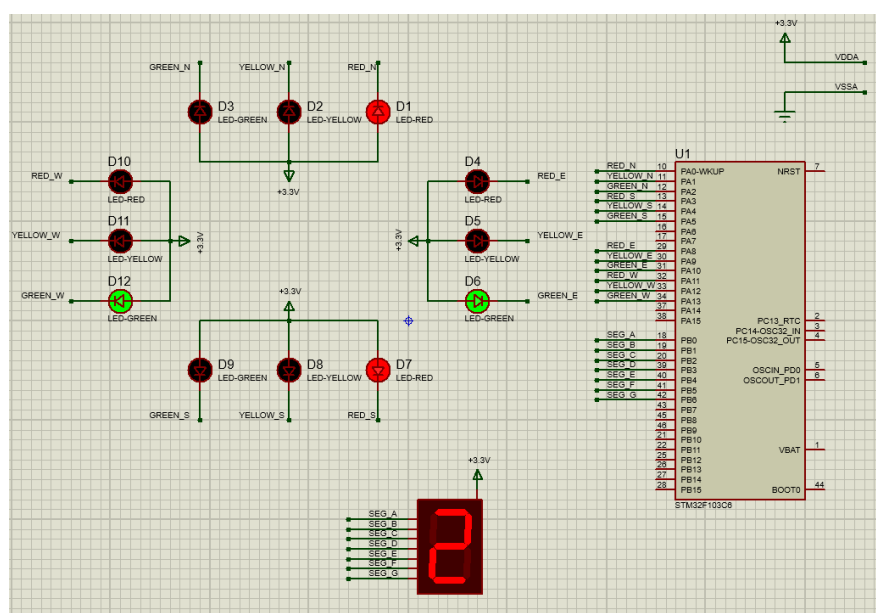
Integrate the 7SEG-LED to the 4 way traffic light. In this case, the 7SEG-LED is used to display countdown value.

In this exercise, only source code is required to present. The function display7SEG in previous exercise can be re-used.



Hình 1.28: Reference design for a 4 way traffic light

Report 1: Present the schematic.



Hình 1.29: Exercise 3 & Exercise 5 Proteus project file [github link](#)

Report 2: Present the source code in while.

```
1 typedef enum {
2     STATE_NS_GREEN,    // North-South green
3     STATE_NS_YELLOW,   // North-South yellow
4     STATE_EW_GREEN,    // East-West green
5     STATE_EW_YELLOW    // East-West yellow
6 } traffic_state_t;
7 static const uint32_t time_green_s = 3;
8 static const uint32_t time_yellow_s = 2;
9 static const uint8_t seg_code[10] = {
10     0xC0, // 0
11     0xF9, // 1
12     0xA4, // 2
13     0xB0, // 3
14     0x99, // 4
15     0x92, // 5
16     0x82, // 6
17     0xF8, // 7
18     0x80, // 8
19     0x90  // 9
20 };
21 // Turn ON/OFF LEDs for one direction
22 static void setTrafficLight(GPIO_TypeDef* RED_Port,
23     uint16_t RED_Pin, GPIO_TypeDef* YELLOW_Port,
24     uint16_t YELLOW_Pin, GPIO_TypeDef* GREEN_Port,
25     uint16_t GREEN_Pin, GPIO_PinState red,
26     GPIO_PinState yellow, GPIO_PinState green)
27 {
28     HAL_GPIO_WritePin(RED_Port, RED_Pin, red);
29     HAL_GPIO_WritePin(YELLOW_Port, YELLOW_Pin, yellow);
30     HAL_GPIO_WritePin(GREEN_Port, GREEN_Pin, green);
31 }
32 // Show a number 0-9 on 7-segment
33 static void display7SEG(int num) {
34     if (num < 0 || num > 9) return;
35
36     uint8_t pattern = seg_code[num];
37
38     HAL_GPIO_WritePin(SEG_A_GPIO_Port, SEG_A_Pin, (pattern
39 & (1<<0)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
40     HAL_GPIO_WritePin(SEG_B_GPIO_Port, SEG_B_Pin, (pattern
41 & (1<<1)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
42     HAL_GPIO_WritePin(SEG_C_GPIO_Port, SEG_C_Pin, (pattern
43 & (1<<2)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
44     HAL_GPIO_WritePin(SEG_D_GPIO_Port, SEG_D_Pin, (pattern
45 & (1<<3)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
46     HAL_GPIO_WritePin(SEG_E_GPIO_Port, SEG_E_Pin, (pattern
47 & (1<<4)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
48     HAL_GPIO_WritePin(SEG_F_GPIO_Port, SEG_F_Pin, (pattern
```

```

43     & (1<<5)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
44     HAL_GPIO_WritePin(SEG_G_GPIO_Port, SEG_G_Pin, (pattern
45     & (1<<6)) ? GPIO_PIN_SET : GPIO_PIN_RESET);
46 }
47 // Init lights: NS green, EW red
48 static void traffic_light_init(void)
49 {
50     // NS: RED off, YELLOW off, GREEN on
51     setTrafficLight(LED_RED_N_GPIO_Port, LED_RED_N_Pin,
52                     LED_YELLOW_N_GPIO_Port, LED_YELLOW_N_Pin,
53                     LED_GREEN_N_GPIO_Port, LED_GREEN_N_Pin,
54                     1, 1, 0);
55     setTrafficLight(LED_RED_S_GPIO_Port, LED_RED_S_Pin,
56                     LED_YELLOW_S_GPIO_Port, LED_YELLOW_S_Pin,
57                     LED_GREEN_S_GPIO_Port, LED_GREEN_S_Pin,
58                     1, 1, 0);
59
60     // EW: RED on, YELLOW off, GREEN off
61     setTrafficLight(LED_RED_E_GPIO_Port, LED_RED_E_Pin,
62                     LED_YELLOW_E_GPIO_Port, LED_YELLOW_E_Pin,
63                     LED_GREEN_E_GPIO_Port, LED_GREEN_E_Pin,
64                     0, 1, 1);
65     setTrafficLight(LED_RED_W_GPIO_Port, LED_RED_W_Pin,
66                     LED_YELLOW_W_GPIO_Port, LED_YELLOW_W_Pin,
67                     LED_GREEN_W_GPIO_Port, LED_GREEN_W_Pin,
68                     0, 1, 1);
69 }
70 // One loop of traffic light
71 static void traffic_light_loop_once(void)
72 {
73     static traffic_state_t state = STATE_NS_GREEN;
74     static int countdown = time_green_s;
75
76     if (countdown <= 0)
77     {
78         switch (state)
79         {
80             case STATE_NS_GREEN: // NS green -> NS yellow
81                 setTrafficLight(LED_RED_N_GPIO_Port, LED_RED_N_Pin,
82                                 LED_YELLOW_N_GPIO_Port, LED_YELLOW_N_Pin,
83                                 LED_GREEN_N_GPIO_Port, LED_GREEN_N_Pin,
84                                 1, 0, 1);
85                 setTrafficLight(LED_RED_S_GPIO_Port, LED_RED_S_Pin,
86                                 LED_YELLOW_S_GPIO_Port, LED_YELLOW_S_Pin,
87                                 LED_GREEN_S_GPIO_Port, LED_GREEN_S_Pin,
88                                 1, 0, 1);
89                 setTrafficLight(LED_RED_E_GPIO_Port, LED_RED_E_Pin,
90                                 LED_YELLOW_E_GPIO_Port, LED_YELLOW_E_Pin,
91                                 LED_GREEN_E_GPIO_Port, LED_GREEN_E_Pin,

```

```

90         0, 1, 1);
91     setTrafficLight(LED_RED_W_GPIO_Port, LED_RED_W_Pin,
92                     LED_YELLOW_W_GPIO_Port, LED_YELLOW_W_Pin,
93                     LED_GREEN_W_GPIO_Port, LED_GREEN_W_Pin,
94                     0, 1, 1);
95
96     state = STATE_NS_YELLOW;
97     countdown = time_yellow_s;
98     break;
99
100    case STATE_NS_YELLOW: // NS yellow -> EW green
101        setTrafficLight(LED_RED_E_GPIO_Port, LED_RED_E_Pin,
102                        LED_YELLOW_E_GPIO_Port, LED_YELLOW_E_Pin,
103                        LED_GREEN_E_GPIO_Port, LED_GREEN_E_Pin,
104                        1, 1, 0);
105        setTrafficLight(LED_RED_W_GPIO_Port, LED_RED_W_Pin,
106                        LED_YELLOW_W_GPIO_Port, LED_YELLOW_W_Pin,
107                        LED_GREEN_W_GPIO_Port, LED_GREEN_W_Pin,
108                        1, 1, 0);
109        setTrafficLight(LED_RED_N_GPIO_Port, LED_RED_N_Pin,
110                        LED_YELLOW_N_GPIO_Port, LED_YELLOW_N_Pin,
111                        LED_GREEN_N_GPIO_Port, LED_GREEN_N_Pin,
112                        0, 1, 1);
113        setTrafficLight(LED_RED_S_GPIO_Port, LED_RED_S_Pin,
114                        LED_YELLOW_S_GPIO_Port, LED_YELLOW_S_Pin,
115                        LED_GREEN_S_GPIO_Port, LED_GREEN_S_Pin,
116                        0, 1, 1);
117
118        state = STATE_EW_GREEN;
119        countdown = time_green_s;
120        break;
121
122    case STATE_EW_GREEN: // EW green -> EW yellow
123        setTrafficLight(LED_RED_E_GPIO_Port, LED_RED_E_Pin,
124                        LED_YELLOW_E_GPIO_Port, LED_YELLOW_E_Pin,
125                        LED_GREEN_E_GPIO_Port, LED_GREEN_E_Pin,
126                        1, 0, 1);
127        setTrafficLight(LED_RED_W_GPIO_Port, LED_RED_W_Pin,
128                        LED_YELLOW_W_GPIO_Port, LED_YELLOW_W_Pin,
129                        LED_GREEN_W_GPIO_Port, LED_GREEN_W_Pin,
130                        1, 0, 1);
131        setTrafficLight(LED_RED_N_GPIO_Port, LED_RED_N_Pin,
132                        LED_YELLOW_N_GPIO_Port, LED_YELLOW_N_Pin,
133                        LED_GREEN_N_GPIO_Port, LED_GREEN_N_Pin,
134                        0, 1, 1);
135        setTrafficLight(LED_RED_S_GPIO_Port, LED_RED_S_Pin,
136                        LED_YELLOW_S_GPIO_Port, LED_YELLOW_S_Pin,
137                        LED_GREEN_S_GPIO_Port, LED_GREEN_S_Pin,
138                        0, 1, 1);

```

```

139
140     state = STATE_EW_YELLOW;
141     countdown = time_yellow_s;
142     break;
143
144     case STATE_EW_YELLOW: // EW yellow -> NS green
145         setTrafficLight(LED_RED_N_GPIO_Port, LED_RED_N_Pin,
146                         LED_YELLOW_N_GPIO_Port, LED_YELLOW_N_Pin,
147                         LED_GREEN_N_GPIO_Port, LED_GREEN_N_Pin,
148                         1, 1, 0);
149         setTrafficLight(LED_RED_S_GPIO_Port, LED_RED_S_Pin,
150                         LED_YELLOW_S_GPIO_Port, LED_YELLOW_S_Pin,
151                         LED_GREEN_S_GPIO_Port, LED_GREEN_S_Pin,
152                         1, 1, 0);
153         setTrafficLight(LED_RED_E_GPIO_Port, LED_RED_E_Pin,
154                         LED_YELLOW_E_GPIO_Port, LED_YELLOW_E_Pin,
155                         LED_GREEN_E_GPIO_Port, LED_GREEN_E_Pin,
156                         0, 1, 1);
157         setTrafficLight(LED_RED_W_GPIO_Port, LED_RED_W_Pin,
158                         LED_YELLOW_W_GPIO_Port, LED_YELLOW_W_Pin,
159                         LED_GREEN_W_GPIO_Port, LED_GREEN_W_Pin,
160                         0, 1, 1);
161
162         state = STATE_NS_GREEN;
163         countdown = time_green_s;
164         break;
165     }
166 }
167 display7SEG(countdown);
168 HAL_Delay(1000);
169 countdown--;
170 }
171 int main(void)
172 {
173     HAL_Init();
174     SystemClock_Config();
175     MX_GPIO_Init();
176
177     traffic_light_init();
178     while (1)
179     {
180         traffic_light_loop_once();
181     }
182 }

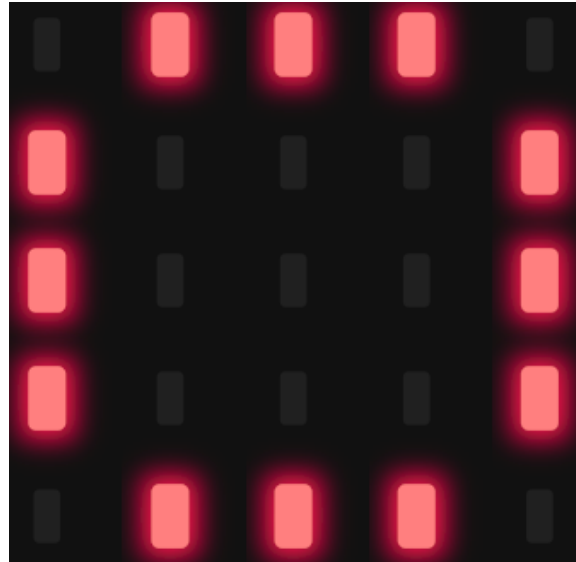
```

Program 1.7: [Exercise 3 & Exercise 5 Source github link](#)

Demo video: [Exercise 3 & Exercise 5 Demo video link](#) (beginning at 01:05).

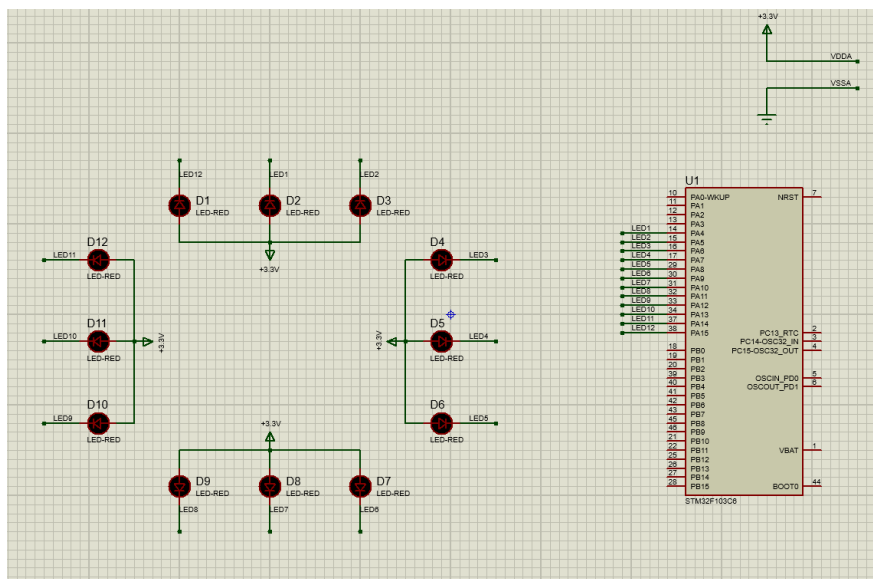
4.5 Exercise 6

In this exercise, a new Proteus schematic is designed to simulate an analog clock, with 12 different number. The connections for 12 LEDs are supposed from PA4 to PA15 of the STM32. The arrangement of 12 LEDs is depicted as follows.



Hình 1.30: 12 LEDs for an analog clock

Report 1: Present the schematic.



Hình 1.31: Exersice 6 Proteus project file [github link](#)

Report 2: Implement a simple program to test the connection of every single LED. This testing program should turn every LED in a sequence.

```
1 static GPIO_TypeDef * const LED_PORT[12] = {  
2     LED0_GPIO_Port, LED1_GPIO_Port, LED2_GPIO_Port,  
    LED3_GPIO_Port,
```

```

3     LED4_GPIO_Port, LED5_GPIO_Port, LED6_GPIO_Port,
      LED7_GPIO_Port,
4     LED8_GPIO_Port, LED9_GPIO_Port, LED10_GPIO_Port,
      LED11_GPIO_Port
5 };
6 static const uint16_t LED_PIN_[12] = {
7     LED0_Pin, LED1_Pin, LED2_Pin, LED3_Pin,
8     LED4_Pin, LED5_Pin, LED6_Pin, LED7_Pin,
9     LED8_Pin, LED9_Pin, LED10_Pin, LED11_Pin
10 };
11 static void led_blink(int led_index, uint32_t delay_ms)
12 {
13     if (led_index >= 12) return;
14     HAL_GPIO_WritePin(LED_PORT[led_index],
15     LED_PIN_[led_index], RESET);
16     HAL_Delay(delay_ms);
17     HAL_GPIO_WritePin(LED_PORT[led_index],
18     LED_PIN_[led_index], SET);
19 }
20
21
22 while (1)
23 {
24     for(int i = 0; i <= 11; i++)
25     {
26         led_blink(i, 500);
27     }
28 }

```

Program 1.8: [Exercise 6 Source github link](#)

Demo video: [Exercise 6 Demo video link](#) (beginning at 02:15).

4.6 Exercise 7

Implement a function named **clearAllClock()** to turn off all 12 LEDs. Present the source code of this function.

```

1 void clearAllClock(){
2     //TODO
3     for (uint8_t i = 0; i < 12; i++) {
4         HAL_GPIO_WritePin(LED_PORT[i], LED_PIN_[i],
5         GPIO_PIN_SET); // OFF
6     }
7 }

```

Program 1.9: [Exercise 7 Source github link](#)

Demo video: [Exercise 7 Demo video link](#) (beginning at 02:47).

4.7 Exercise 8

Implement a function named **setNumberOnClock(int num)**. The input for this function is from **0 to 11** and an appropriate LED is turn on. Present the source code of this function.

```
1 void clearAllClock(){
2     //TODO
3     for (uint8_t i = 0; i < 12; i++) {
4         HAL_GPIO_WritePin(LED_PORT[i], LED_PIN_[i],
5         GPIO_PIN_SET); // OFF
6     }
7 }
```

Program 1.10: [Exercise 8 Source github link](#)

Demo video: [Exercise 8 Demo video link](#) (beginning at 03:04).

4.8 Exercise 9

Implement a function named **clearNumberOnClock(int num)**. The input for this function is from **0 to 11** and an appropriate LED is turn off.

```
1 static void clearNumberOnClock(int num)
2 {
3     if (num < 0 || num > 11) return;
4     HAL_GPIO_WritePin(LED_PORT[num], LED_PIN_[num],
5     GPIO_PIN_SET); // Active-low: RESET = turn ON, SET =
6     turn OFF
7 }
```

Program 1.11: [Exercise 9 Source github link](#)

Demo video: [Exercise 9 Demo video link](#) (beginning at 03:33).

4.9 Exercise 10

Integrate the whole system and use 12 LEDs to display a clock. At a given time, there are only 3 LEDs are turn on for hour, minute and second information.

```
1 static GPIO_TypeDef * const LED_PORT[12] = {
2     LED0_GPIO_Port, LED1_GPIO_Port, LED2_GPIO_Port,
3     LED3_GPIO_Port, LED4_GPIO_Port, LED5_GPIO_Port,
4     LED6_GPIO_Port, LED7_GPIO_Port, LED8_GPIO_Port,
5     LED9_GPIO_Port, LED10_GPIO_Port, LED11_GPIO_Port
6 };
7 static const uint16_t LED_PIN_[12] = {
8     LED0_Pin, LED1_Pin, LED2_Pin, LED3_Pin,
9     LED4_Pin, LED5_Pin, LED6_Pin, LED7_Pin,
10    LED8_Pin, LED9_Pin, LED10_Pin, LED11_Pin
11 };
```

```

12
13 static void clearAllClock(void)
14 {
15     for (uint8_t i = 0; i < 12; i++) {
16         HAL_GPIO_WritePin(LED_PORT[i], LED_PIN_[i],
17         GPIO_PIN_SET); // OFF
18     }
19 }
20 static void setNumberOnClock(int num)
21 {
22     if (num < 0 || num > 11) return;
23     HAL_GPIO_WritePin(LED_PORT[num], LED_PIN_[num],
24     GPIO_PIN_RESET);
25 }
26 void displayClock(int hour, int minute, int second)
27 {
28     clearAllClock();
29
30     int h = hour % 12;
31     int m = minute / 5;
32     int s = second / 5;
33
34     setNumberOnClock(h);
35     setNumberOnClock(m);
36     setNumberOnClock(s);
37 }
38
39 int main(void)
40 {
41     HAL_Init();
42     SystemClock_Config();
43     MX_GPIO_Init();
44
45     clearAllClock();
46     uint8_t hour=1, minute=50, second=0;
47     while (1)
48     {
49         displayClock(hour, minute, second);
50         HAL_Delay(20);
51         second++;
52         if(second>=60) {
53             minute++;
54             second=0;
55         }
56         if(minute>=60) {
57             hour++;
58             minute=0;
59         }
60         if(hour>=24){

```

```
59     hour=0;  
60     }  
61     }  
62 }
```

Program 1.12: [Exercise 10 Source github link](#)

Demo video: [Exercise 10 Demo video link](#) (beginning at 04:02).